



Artificial Intelligence & Autonomy

The Halo System and VaxCore Platform

Voltaris Mobility N.V. • Amsterdam, Netherlands • Founded 2016

“Drive the Current.”



Intelligence at the Core

Artificial intelligence is not an add-on at Voltaris; it is the foundation of the company's product strategy. Every vehicle ships with Halo, the company's driver-assistance and autonomy system, running on a custom in-house compute platform called VaxCore. The long-term goal is fully autonomous transport — cars that can drive themselves both for private owners and as part of the Voltaris Go robotaxi fleet.

The Halo System

Halo is a full-stack autonomous-driving system that combines perception, prediction, planning and control. It fuses data from cameras, radar and lidar to build a real-time model of the world around the vehicle, predicts the behaviour of other road users, and plans a safe path. Halo is delivered in tiers, allowing customers to start with assisted driving and upgrade over the air to higher levels of autonomy.

Halo Tier	Capability	Availability
Halo Assist	Adaptive cruise, lane-keeping, parking	Standard on all models
Halo Pilot	Hands-off highway driving (supervised)	Optional on Aero, Atlas, Orion
Halo Autonomy	Level 4 self-driving in mapped zones	Helix fleet; select cities

The VaxCore Compute Platform

VaxCore is Voltaris' purpose-designed automotive AI computer. It runs the neural networks behind Halo with redundant processing for safety, and is powerful enough to handle the sensor load of the fully autonomous Helix. Designing the chip in-house lets Voltaris tune hardware and software together and push improvements to the entire fleet through over-the-air updates.

Levels of Autonomy

Voltaris describes its progress using the standard SAE framework. Today, the consumer fleet operates at Level 2+ (assisted, with the driver responsible) through Halo Pilot, while the Helix robotaxi operates at Level 4 (no human driver required) within carefully mapped and geofenced city zones. The company's roadmap targets broader Level 4 availability city-by-city as mapping, regulation and validation allow.

Data, Learning & Safety

Halo improves through a continuous learning loop: anonymised data from the fleet surfaces rare and difficult driving situations, which are used to retrain the models, validate them in simulation, and deploy improved versions over the air. Voltaris emphasises that safety drives every deployment decision, and that the system is only allowed to operate autonomously where it has been extensively validated.

- Sensor fusion across cameras, radar and lidar.



- Custom in-house VaxCore compute with safety redundancy.
- Over-the-air updates improve every vehicle in the field.
- Simulation-first validation before any real-world deployment.
- Level 4 autonomy live today in geofenced robotaxi zones.